

Task - 1

Date: _____

- 1) Class C
- 2) Class A
- 3) Class B
- 4) Class D
- 5) Class A

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Task 2

- 1) invalid , 300 is greater than 255
- 2) invalid , 256 is greater than 255
- 3) valid
- 4) valid
- 5) valid

Task 3

ipconfig ← command

ip → 172.16.204.40

Task 4

Traditionally switch to switch requires a crossover cable and Router to PC requires a crossover cable.

However, modern devices with Auto-MDI can often use a straight-through cable for either connection.

Task 5

In a normal home installation, I would use a Cat 5e or Cat 6 UTP straight-through ethernet cable to connect computers and other devices to the router or switch. Modern networking devices generally support Auto MDI so crossover cables are usually unnecessary.

Task 6

The IP connectivity is working because the computer can successfully ping 192.168.0.2. However, the hostname "blue machine" cannot be resolved to an IP ~~machine~~ address. The problem is therefore likely related to DNS, Net BIOS, hostname configuration, or local name resolution rather than the physical network connection.

Task 7

A typical home network uses a star topology where a central router connects devices such as

Command Prompt

Microsoft Windows [Version 10.0.19045.6456]
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C:\Users\k240851>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet 2:

Connection-specific DNS Suffix . :
Link-local IPv6 Address : fe80::8c0b:8494:1f96:b1c1%17
IPv4 Address. : 192.168.56.1
Subnet Mask : 255.255.255.0
Default Gateway :

Ethernet adapter Ethernet 3:

Connection-specific DNS Suffix . : khifast.nu.edu.pk
Link-local IPv6 Address : fe80::bd9:3f7b:4ef4:a3ff%18
IPv4 Address. : 172.16.204.40
Subnet Mask : 255.255.255.0
Default Gateway : 172.16.204.1

Ethernet adapter VMware Network Adapter VMnet1:

Connection-specific DNS Suffix . :
Link-local IPv6 Address : fe80::a06b:4a81:a643:ce41%8
IPv4 Address. : 192.168.232.1
Subnet Mask : 255.255.255.0
Default Gateway :

Ethernet adapter VMware Network Adapter VMnet8:

Connection-specific DNS Suffix . :
Link-local IPv6 Address : fe80::d997:2d96:7118:9d9%4
IPv4 Address. : 192.168.145.1
Subnet Mask : 255.255.255.0
Default Gateway :

C:\Users\k240851>

